

BSCS FINAL PROJECT

Requirements Specification

ReMIND: Generative AI based Memory Rebuilder



Project Advisor

Muhammad Zulkifl Hasan

Presented by:

Group ID: F25CS104

Registration No:

L1F22BSCS0086

L1F22BSCS0103

L1F22BSCS0094

Name:

M. Hassan Ashraf

M. Talha Faizan

Mubashar Tanveer

Faculty of Information Technology & Computer Science

University of Central Punjab

Software Requirements Specification

Version 1

ReMIND: Generative AI based Memory Rebuilder

Advisor: Muhammad Zulkifl Hasan

Group F25CS104

Member Name	Primary Responsibility
M Hassan Ashraf	Design and Strategy
M Talha Faizan	Frontend Development
Mubashar Tanveer	Documentation and Backend

Table of Contents:

Revision History	ii
Previous Phases Feedback.....	ii
Abstract.....	iii
1. Introduction and Background	1
1.1 Product (Problem Statement)	1
1.2 Background	1
1.3 Scope	1
1.4 Objective(s)/Aim(s)/Target(s)	1
1.5 Challenges	2
1.6 Learning Outcomes	2
1.7 Nature of End Product.....	3
1.8 Completeness Criteria	3
1.9 Business Goals	3
1.10 Related Work/ Literature Survey/ Literature Review	3
1.11 Document Conventions.....	4
2. Overall Description	4
2.1 Product Features.....	4
2.2 User Classes and Characteristics.....	5
2.3 Operating Environment.....	5
2.4 Design and Implementation Constraints	6
2.5 Assumptions and Dependencies.....	6
3. Functional Requirements	7
3.1 Upload Multimedia Input	7
3.2 Generate Reconstructed Memory.....	7
3.3 Manage Users and System Data.....	8
3.4 Requirements Analysis and Modeling	9
4. Nonfunctional Requirements	11
4.1 Performance Requirements	11
4.2 Safety Requirements	11
4.3 Security Requirements	11
4.4 Additional Software Quality Attributes	11
5. Other Requirements	12
6. Revised Project Plan	12
7. References.....	12
Appendix A: Glossary.....	13
Appendix B: IV & V Report	14

Revision History

Name	Date	Reason For Changes	Version
Idea Defense	September 11, 2025	Problem statements and scope are not clear.	Version 2

Previous Phases Feedback

Idea Defense Feedback (Screenshot)

||

Abstract

ReMIND is an AI assisted web application designed to support individuals experiencing memory loss disorders such as Alzheimer’s disease, dementia, and age-related cognitive decline. Memory loss affects not only a person’s ability to recall past events but also their emotional stability, relationships, and overall sense of identity. Most existing digital solutions rely heavily on manual input, and when no prior record or recollection exists, they fail to assist patients and caregivers in restoring meaningful experiences. ReMIND addresses this limitation by offering semi-automated memory reconstruction that helps users reconnect with their personal history.

The system collects and analyzes different forms of data, including written notes, photographs, and voice recordings, and organizes them into personalized timelines that reflect moments from daily life. Through artificial intelligence, ReMIND generates contextual details that can suggest likely places, people, and events related to the user’s memories. This enriched information helps patients retrieve forgotten experiences in a more natural and emotionally resonant way. The platform creates an interactive environment where patients and caregivers can revisit, discuss, and refine reconstructed memories together, strengthening emotional connection and supporting cognitive engagement.

The project brings together concepts from artificial intelligence, natural language processing, computer vision, speech processing, data management, and human centered design. The focus is not solely on automation but on creating an experience that prioritizes empathy, accessibility, and user comfort. By transforming fragmented or incomplete data into meaningful memory narratives, ReMIND aims to restore a sense of identity and continuity for individuals facing cognitive decline.

The expected outcome is a functional prototype that demonstrates how intelligently guided memory reconstruction can contribute to cognitive rehabilitation and long-term digital memory preservation. ReMIND highlights the potential of emerging AI systems to support emotional wellbeing and provide hope for individuals whose memories are fading.

1. Introduction and Background

ReMIND is an AI-assisted web application aimed at helping individuals with memory loss, such as Alzheimer's and dementia patients, relive past experiences. It reconstructs fragmented memories using multimedia inputs. The project is motivated by the emotional and cognitive challenges faced by patients, emphasizing empathy, recall, and caregiver support through technology.

1.1 Product (Problem Statement)

ReMIND is an intelligent, web-based memory reconstruction system designed to assist individuals suffering from memory loss disorders such as Alzheimer's and dementia. The software addresses the limitations of existing reminder tools that rely heavily on manual event logging or user recall, which often fail when sufficient details are unavailable. ReMIND integrates multiple data sources such as photos, text notes, and voice recordings to automatically identify, organize, and recreate meaningful events. Using artificial intelligence and contextual analysis, the system reconstructs daily timelines and generates narrative memories that patients and caregivers can revisit. Its user-friendly, elderly-oriented interface ensures accessibility and comfort, while secure backend services manage data storage and processing. The system provides collaborative features for caregivers, enabling them to contribute and review reconstructed memories. Ultimately, ReMIND bridges emotional and cognitive gaps, improving quality of life by transforming fragmented inputs into coherent, emotionally engaging experiences that support recall and human connection.

1.2 Background

The project lies within the domains of artificial intelligence, cognitive rehabilitation, and healthcare technology. It focuses on applying AI techniques such as natural language processing, computer vision, and speech processing to assist individuals with memory loss. By merging technology and empathy, ReMIND enhances memory recall, emotional well-being, and caregiver collaboration.

1.3 Scope

The scope of ReMIND covers the design and development of a web-based application that reconstructs and presents memories using multimedia inputs such as text, images, and audio. The system includes event clustering, memory timeline generation, and a reminder module for patients and caregivers. It focuses on enhancing recall and emotional connection through AI-powered contextual analysis. The application will feature an elderly-friendly interface, secure data handling, and cloud-based deployment. Medical diagnosis, hospital EHR integration, and real-time biometric tracking are excluded from the project. It will be a Concept-of-Proof limited project.

1.4 Objective(s)/Aim(s)/Target(s)

The primary objective of ReMIND is to develop an AI-assisted web application that helps individuals with memory loss to reconstruct meaningful memories from fragmented multimedia inputs. The project aims to combine technology and empathy to create a system that supports both patients and caregivers in recalling and preserving life experiences.

ReMIND: Generative AI based Memory Rebuilder

The specific objectives are as follows:

Memory Reconstruction: Transform incomplete data such as text notes, photos, and voice recordings into coherent and contextually rich memory narratives.

Event Clustering: Use AI techniques to organize related media and information into structured timelines representing daily or significant events.

Reminder and Notification Module: Provide timely reminders and alerts to assist patients in maintaining routine and connection with daily life.

Elderly-Friendly Interface: Design a simple, accessible interface optimized for older users and non-technical caregivers.

Secure Data Handling: Implement encryption, authentication, and cloud storage to protect sensitive personal data.

Caregiver Collaboration: Enable caregivers to contribute and review reconstructed memories to ensure accuracy and emotional value.

Proof of Concept Demonstration: Deliver a functional prototype using existing AI APIs and web technologies to validate the concept.

Overall, ReMIND seeks to enhance emotional well-being, cognitive support, and human connection through accessible AI innovation.

1.5 Challenges

Developing ReMIND presents several technical and practical challenges. Integrating multimodal data such as text, audio, and images into a coherent memory reconstruction pipeline requires effective synchronization and contextual understanding. Ensuring reliable performance using limited computing resources adds complexity, especially for AI processing. Data privacy and ethical handling of personal memories are critical concerns. Designing an intuitive, elderly-friendly interface demands careful attention to accessibility and simplicity. Seamlessly integrating third-party AI APIs, such as Google Vision, Whisper, and ElevenLabs, while maintaining performance consistency also poses challenges. Additionally, validating reconstructed memories for accuracy and emotional relevance remains a significant research and design challenge.

1.6 Learning Outcomes

Through the development of ReMIND, we will gain hands-on experience in applying artificial intelligence, natural language processing, computer vision, and speech processing to solve real-world healthcare challenges. Also, we will learn to integrate multiple AI APIs and manage communication between frontend and backend systems using React.js and Flask/FastAPI. The project will strengthen skills in database design, secure data handling, and cloud deployment. We will also develop a deeper understanding of user-centered design, especially for accessibility and elderly usability. Overall, the

project enhances both technical proficiency and awareness of how technology can address human and emotional needs.

1.7 Nature of End Product

The final product of ReMIND will be a fully functional, AI-assisted web application that reconstructs fragmented memories using multimedia inputs such as text, images, and audio. It will feature an interactive chat-like interface where users can view, listen to, or read about reconstructed events. The system will include integrated AI modules for image enhancement, speech transcription, and text generation, supported by a secure backend and database. The application will serve as a proof-of-concept demonstrating the potential of AI in cognitive rehabilitation, emphasizing emotional engagement, accessibility, and human-centered design for patients with memory loss and their caregivers.

1.8 Completeness Criteria

The project will be considered complete when the ReMIND web application successfully reconstructs memories from multimedia inputs, generates coherent timelines, and allows patient–caregiver interaction through a functional, user-friendly interface. Integration of AI APIs, secure data handling, and successful deployment with documented testing and evaluation will mark project completion.

Sr No.	Criteria	Weightage %
1	Frontend Development (React.js UI/UX)	15%
2	Backend Development (Flask/FastAPI, Database)	20%
3	AI Integration (via APIs)	30%
4	Reminder & Notification Module	10%
5	Storage & Deployment	10%
6	Testing & Evaluation	15%

1.9 Business Goals

As a proof-of-concept prototype, ReMIND aims to demonstrate the commercial potential of AI-driven memory reconstruction in healthcare and assisted living. Its primary goal is to validate the feasibility of using artificial intelligence for emotional and cognitive support in patients with memory loss. The project establishes a foundation for future development into a scalable product that could benefit healthcare institutions, caregivers, and families. By integrating accessible technology with empathetic design, ReMIND highlights opportunities for collaboration between AI developers and healthcare providers, showcasing how innovation can lead to meaningful, socially impactful digital health solutions in memory rehabilitation.

1.10 Related Work/ Literature Survey/ Literature Review

The increasing prevalence of Alzheimer’s and dementia has led to a growing demand for intelligent memory-assistive systems that go beyond traditional reminders. Existing digital tools often rely solely on manual logging or event notes, which fail when patients cannot recall sufficient information. According to Alzheimer’s Disease International, over 55 million people live with dementia worldwide, emphasizing the urgent need for supportive, technology-driven solutions [\[6\]](#).

ReMIND: Generative AI based Memory Rebuilder

Several AI-driven platforms and research projects have demonstrated potential in isolated aspects of memory reconstruction. MyHeritage's *Deep Nostalgia* introduced facial animation in static photos, highlighting emotional engagement through visual recall [3]. Voice technologies such as ElevenLabs and Descript provide realistic voice synthesis and cloning, enabling the recreation of familiar voices for emotional resonance [4], [5]. Real-ESRGAN, an open-source image restoration model, improves photo clarity, making old or low-quality images usable in memory visualization [1]. Meanwhile, OpenAI's Whisper model delivers robust speech-to-text capabilities, ideal for transcribing fragmented audio memories [2].

While these systems excel individually, few integrate multimodal data text, audio, and images into a unified cognitive support framework. ReMIND distinguishes itself by combining these technologies into an accessible web platform, allowing patients and caregivers to collaboratively reconstruct meaningful, context-aware memory timelines enhanced by AI-generated narratives and multimedia synthesis [7].

1.11 Document Conventions

The Document Conventions being used throughout the document are following:

- Text being used is Times New Roman.
- Main Headings are of Size 16.
- Sub-Headings are of Size 14.
- Content/Paragraph and Tertiary headings are of Size 12.
- Auto line spacing is used.

2. Overall Description

2.1 Product Features

ReMIND is designed as an AI-powered, web-based memory reconstruction system that integrates multimedia data text, images, and audio to help individuals with memory loss to relive and preserve meaningful life events. The product emphasizes accessibility, emotional relevance, and technological feasibility.

Key features include:

Memory Reconstruction Engine: Uses AI models to analyze multimedia inputs and generate coherent, context-rich narratives that rebuild fragmented memories.

Event Clustering and Timeline Generation: Automatically organizes related media and data into daily or event-based timelines, enabling users to view reconstructed memories chronologically.

Multimedia Enhancement: Employs tools such as Real-ESRGAN for image enhancement, Whisper for voice transcription, and ElevenLabs for realistic voice synthesis.

Interactive Chat Interface: Provides a user-friendly conversational interface where patients and caregivers can interact, view, and refine reconstructed memories.

Reminder and Notification Module: Assists patients with daily activities and event tracking through customizable reminders.

Caregiver Collaboration: Allows caregivers to contribute to additional context or verify memory accuracy.

Secure Cloud Storage: Ensures privacy and safety through encrypted data handling and cloud-based management.

Overall, ReMIND combines artificial intelligence, empathy, and intuitive design to transform scattered personal data into emotionally significant digital experiences that aid cognitive rehabilitation and emotional well-being.

2.2 User Classes and Characteristics

ReMIND is designed to serve two primary user classes **Patients (Memory Loss Individuals)** and **Caregivers (Family Members or Healthcare Assistants)** along with two secondary user classes System Administrators and Researchers/Developers. Each user class interacts with the system at different levels of complexity, privilege, and frequency.

1. **Patients** (Primary Users):
These are individuals affected by Alzheimer's, dementia, or general memory impairment. They use ReMIND frequently for viewing, listening to, and interacting with reconstructed memories or receiving reminders. The interface is designed to be elderly-friendly, visually clear, and minimal in complexity. Accessibility features such as large fonts, audio narration, and simplified navigation are prioritized.
2. **Caregivers** (Primary Users):
Family members, nurses, or attendants assist patients in managing and verifying reconstructed memories. They have moderate technical expertise and access privileges that allow them to upload multimedia inputs, annotate events, and review memory accuracy. Their primary goal is to monitor progress, reduce patient stress, and maintain emotional connections.
3. **System Administrators** (Secondary Users):
Administrators manage backend operations, including user account verification, data security, and cloud services. They ensure system reliability and compliance with ethical data standards.
4. **Researchers/Developers** (Secondary Users):
They may use ReMIND as a testbed to explore improvements in AI-based cognitive rehabilitation. Their access focuses on algorithmic evaluation and performance monitoring.

Favored user classes are **patients** and **caregivers**, as the system is primarily designed around their emotional and cognitive needs, while other classes ensure technical and operational continuity.

2.3 Operating Environment

ReMIND will operate as a web-based application accessible through modern browsers on both desktop and laptop systems. The system is designed to run smoothly on mid-level hardware and within a university or home computing setup.

Client Environment: Users can access ReMIND through standard web browsers such as Google Chrome, Microsoft Edge, or Mozilla Firefox. The front-end interface, built using React.js and CSS3, ensures compatibility across Windows, macOS, and Linux operating systems. A stable internet connection is required for AI-assisted features like speech recognition, memory reconstruction, and image enhancement, which depend on external APIs.

Server Environment: The backend operates on a Flask/FastAPI server using Python, hosted on Render or Heroku. PostgreSQL serves as the primary database for storing user data, media references, and contextual metadata. The backend communicates securely through RESTful APIs.

AI Processing Environment: For AI-based modules such as voice transcription, image restoration, and memory reconstruction, ReMIND leverages APIs from OpenAI (GPT-4, DALL·E 3, Whisper), Stable Diffusion, and ElevenLabs. Intensive processing tasks may be performed using UCP's GPU-enabled lab machines equipped with NVIDIA RTX 4060 GPUs.

Development and Testing Environment: Development will be carried out on personal laptops (Windows 10/11 or Ubuntu 22.04) with Visual Studio Code, GitHub for version control, and PostgreSQL for local database testing. Browser-based testing tools and Postman will verify frontend-backend communication.

2.4 Design and Implementation Constraints

The design and development of ReMIND are influenced by several technical, operational, and practical constraints. Since the project is a prototype developed within limited academic resources, it must operate efficiently on mid-level personal laptops without dedicated GPUs. Internet connectivity is essential for the proper functioning of these services, making the system dependent on stable network availability. The project also adheres to ethical and privacy constraints, as it involves personal and potentially sensitive multimedia data; thus, strict measures for data handling, encryption, and storage must be followed. The application is developed using specific technologies React.js for the frontend, Flask/FastAPI for the backend, and PostgreSQL for database management to maintain consistency and ensure ease of deployment on platforms like Render and Heroku. Because the scope is limited to a proof-of-concept system, real-time synchronization with hospital EHRs or wearable IoT devices is intentionally excluded. The user interface must remain lightweight, accessible, and elderly-friendly, imposing design restrictions on complex animations or heavy data visualization. These combined constraints guide the project's architecture, performance expectations, and implementation feasibility.

2.5 Assumptions and Dependencies

The development and operation of ReMIND are based on several key assumptions and dependencies that influence its design and functionality. It is assumed that users, both patients and caregivers, will have access to stable internet connectivity, as the application relies on cloud-based services and external AI APIs for processing multimedia inputs such as images, voice notes, and text logs. The system assumes the continued availability and reliability of third-party APIs, including OpenAI Whisper, ElevenLabs, Google Vision, and Real-ESRGAN, which are essential for transcription, image enhancement, and voice synthesis. Any change in licensing, API limits, or service accessibility could directly affect system functionality. It is also assumed that UCP's GPU-equipped laboratory systems will remain accessible for heavy AI processing and testing when required. Additionally, PostgreSQL, Firebase, and Cloudinary services are assumed to remain stable for secure storage and retrieval of multimedia data. User devices are expected to support modern browsers compatible with React.js and JavaScript ES6 standards. Finally, it is assumed that ethical and privacy guidelines will be followed by all team members and that sample datasets used during testing will not contain personally identifiable information.

3. Functional Requirements

3.1 Upload Multimedia Input

Identifier		Multimedia Uploading
Purpose		To allow users to upload photos, audio, or text that will be the input for the memory OR to enter text.
Priority		High
Pre-conditions		Users must be logged into the system. The device must have a stable internet connection.
Post-conditions		Uploaded multimedia files are successfully stored in Firebase/Cloudinary, and metadata is recorded in PostgreSQL.
Typical Course of Action		
S#	Actor Action	System Response
1	User logs into the ReMIND web application.	System verifies credentials and loads the dashboard.
2	User selects the Upload Memory option.	System displays upload options for photo, audio, or text.
3	The user confirms the upload.	System stores the file in cloud storage and logs its metadata.
4	System notifies users of successful uploads.	Confirmation message displayed and file becomes available for memory reconstruction.
Alternate Course of Action		
S#	Actor Action	System Response
1	User uploads an invalid or unsupported file.	System displays an error message and prompts re-upload.
2	Internet connection is lost.	System pauses upload and provides retry options once connection is restored.

Table 1: UC-1

3.2 Generate Reconstructed Memory

Identifier		Memory Reconstruction
Purpose		To generate a meaningful event summary by processing multimedia inputs using AI services.
Priority		High

Pre-conditions		Users have uploaded multimedia content. AI APIs are accessible and functional.
Post-conditions		Reconstructed memory is stored and displayed to the user in the timeline.
Typical Course of Action		
S#	Actor Action	System Response
1	User selects uploaded files for reconstruction.	The system retrieves the selected files from storage.
2	User clicks Generate Memory .	The system sends files to integrated APIs (Whisper, Vision, etc.) for analysis.
3	The user waits for the result.	The system processes text, image, and audio data.
4	User reviews the generated memory.	The system displays reconstructed narrative, enhanced image, and optional audio playback.
5	The user saves memory.	The system stores the final output in the database and updates the user's timeline.
Alternate Course of Action		
S#	Actor Action	System Response
1	AI services fail or timeout.	The system shows a retry option and logs the error.
2	Uploaded files are incomplete or corrupted.	The system will tell the user to verify the files.

Table 2: UC

3.3 Manage Users and System Data

Identifier	System Data Management
-------------------	------------------------

ReMIND: Generative AI based Memory Rebuilder

Purpose		To allow the administrator to manage user accounts, monitor system activity, and configure AI services within ReMIND app.
Priority		High
Pre-conditions		Admin must log in with valid credentials and have access privileges.
Post-conditions		All user or system configuration changes are updated in the database.
Typical Course of Action		
S#	Actor Action	System Response
1	Admin logs into the ReMIND	The system verifies credentials and loads the admin dashboard.
2	Admin selects a user account to edit, disable, or remove.	The system validates the action and updates the user's status accordingly.
3	Admin reviews and updates API usage statistics or system logs.	The system fetches the requested logs and updates them.
Alternate Course of Action		
S#	Actor Action	System Response
1	Admin attempts to delete a protected or active user account.	The system denies the action and displays a warning message.
2	Admin session expires during activity.	The system logs out the admin and asks for authentication again.

Table 3: UC-3

3.4 Requirements Analysis and Modeling

Use

Case

Diagrams:

UC – 1:

UC – 2:

UC – 3:

ERD Diagram:

Relational Schema:

output_path

UML Class Diagram:



Data Flow Diagram:

Fig: DFD – Level 0

Fig: DFD – Level 1

Fig: DFD – Level 2

Sequence Diagram:

Decision Table:

Rule	Input	Alternative Conditions	Action	Output
------	-------	------------------------	--------	--------

ReMIND: Generative AI based Memory Rebuilder

R1	User provides valid login credentials	Invalid credentials	Authenticate user, create session	Access granted / error message displayed
R2	Patient uploads multimedia input	Unsupported format or missing metadata	Accept input, store temporarily, trigger preprocessing	Acknowledgment / error message
R3	Uploaded file validated successfully	File corrupted or exceeds size limit	Proceed to AI preprocessing and enhancement	Process initiated / warning prompt
R4	Image input detected	No image detected	Trigger Real-ESRGAN enhancement and tagging	Enhanced image + metadata stored
R5	Audio input detected	No audio provided	Send to Whisper API for transcription	Transcript text returned
R6	Text or transcript available	Missing or empty text input	Send to GPT-4 for contextual understanding	Structured text/context response
R7	All media processed successfully	Any AI module fails	Aggregate data and reconstruct event	Reconstructed memory / retry message
R8	Narrative generation successful	AI fails to generate or timeout	Store reconstructed memory in repository	Text + voice response / system retry
R9	Reminder set by caregiver	Reminder missing or invalid	Save reminder in database	Reminder confirmation / validation error
R10	Feedback submitted by patient/caregiver	Feedback skipped	Store feedback in log	Confirmation message / skip noted
R11	Admin monitors system logs	Logs not accessible	Display system status dashboard	Log summary / permission error

Event Table:

Event	Event Name	Trigger Type	Source	System Activity	Output
E1	User Login	External (User Action)	Patient / Caregiver / Admin	Authenticate credentials, create session	Dashboard or role-based access
E2	Media Upload	External (User Action)	Patient / Caregiver	Validate and store uploaded media	Temporary file storage + confirmation
E3	Data Preprocessing	Internal (System)	Backend	Cleans, formats, validates input data	Ready media for AI modules
E4	Image Enhancement	External (AI Service)	Real-ESRGAN	Enhances image and extracts context	Enhanced image file stored
E5	Audio Transcription	External (AI Service)	Whisper API	Converts audio to text, detects tone	Transcript returned
E6	Text Analysis	External (AI Service)	GPT-4	Interprets and structures text context	Semantic data created
E7	Event Reconstruction	Internal (AI Process)	AI Module	Combines multimodal data into cohesive memory	Structured narrative generated
E8	Voice Synthesis	External (AI Service)	ElevenLabs	Converts narrative text to natural audio	Voice file generated and stored

ReMIND: Generative AI based Memory Rebuilder

E9	Memory Storage	Internal (Database)	Backend → PostgreSQL	Saves reconstructed memory with metadata	Record created in memory repository
E10	Memory Retrieval	External (User Request)	Patient / Caregiver	Fetches requested memory or timeline	Displayed in frontend interface
E11	Reminder Creation	External (User Action)	Caregiver	Adds reminder entry in database	Confirmation sent to user
E12	Reminder Notification	Internal (Timed Event)	Notification Module	Sends scheduled alert to patient	Notification on UI
E13	Feedback Submission	External (User Action)	Patient / Caregiver	Logs ratings and suggestions	Stored in feedback log
E14	System Monitoring	Internal (Admin Action)	Administrator	Reviews logs, checks API integrations	Diagnostic report generated

4. Nonfunctional Requirements

4.1 Performance Requirements

ReMIND is designed to operate as a responsive and efficient web-based application capable of handling multimedia data processing, retrieval, and AI-assisted event reconstruction in real time or near real time. The system should load the main dashboard and associated modules within **3 to 5 seconds** under normal network conditions. Uploading and processing small multimedia files (such as images or short audio clips) should not exceed **10 to 15 seconds**, depending on file size and API response time. The application must be optimized to support at least **50 concurrent users**. Network latency and third-party API response times may affect reconstruction performance. Therefore, caching mechanisms will be used to improve data retrieval and reduce repetitive processing. The front-end interface will ensure smooth navigation and responsiveness across devices. Furthermore, since the project will be tested on mid-level hardware, performance tuning will prioritize lightweight operations and memory-efficient data handling.

4.2 Safety Requirements

ReMIND must ensure that all user data, including text, audio, and images, is protected from loss, corruption, or unauthorized access. To maintain data integrity and safety, all sensitive information stored within the system will be encrypted using **AES-256 encryption**, preventing unauthorized access. At the application level, **OAuth 2.0** and **JWT (JSON Web Token)** will be used to authenticate users securely and verify access privileges, reducing risks of unauthorized operations or misuse. The system will include confirmation prompts before any deletion or modification of important data to avoid accidental loss. Regular data backups and recovery mechanisms will be implemented to safeguard information in case of unexpected failures. These combined safety measures ensure that user data remains secure, accurate, and protected throughout system operation.

4.3 Security Requirements

ReMIND must ensure secure communication, access control, and data privacy for all users, including patients, caregivers, and doctors. The system will use multiple layers of security to prevent unauthorized access, data breaches, or misuse of sensitive information. All communication between the client and server will be protected using **SSL, TLS, HTTPS, SSH, and FTPS** protocols to ensure encrypted data

transmission over the network. Network access control will work with **802.1X** and **WPA3** standards to safeguard against unauthorized devices or connections. User authentication and authorization will follow strict identity verification processes, processed by **OAuth 2.0** and **JWT** for secure API communication and session management. User credentials and tokens will be encrypted before storage and never transmitted in plain text. The system will also include measures to detect unusual login attempts, enforce session timeouts, and restrict access based on user roles (Admin, Doctor, or Patient). ReMIND adheres to international data security and privacy standards, including **ISO 27001**, **SOC 2**, **HIPAA**, and **GDPR**, ensuring that personal and health-related information is handled responsibly and meets global protection requirements. These protocols collectively provide a secure foundation for user trust, data confidentiality, and long-term system reliability.

4.4 Additional Software Quality Attributes

ReMIND focuses on usability, reliability, and maintainability to ensure a smooth user experience for patients, caregivers, and doctors. The system will have at least 99% uptime for consistent availability. The interface is designed to be easy to use and accessible, especially for elderly users, with clear navigation and simple controls. Reliability is ensured through proper error handling, data validation, and recovery options. The system's modular architecture supports easy updates and maintainability, while portability allows access across different devices and browsers. Components and APIs are reusable, making future improvements or integration easier. Regular testing and debugging will maintain correctness and robustness, ensuring that the system performs accurately and remains stable under normal usage conditions.

5. Other Requirements

ReMIND will use PostgreSQL as the primary database for storing user data and Firebase/Cloudinary for multimedia storage. The system requires a stable internet connection for API communication and data synchronization. All external integrations, including AI services like OpenAI, Whisper, and Google Vision, must follow proper API authentication and usage policies. The application will support English as the primary language, with the option for localization in future versions. ReMIND must comply with data privacy and legal standards such as HIPAA and GDPR for handling personal information. Reuse of modular components and APIs will be encouraged for future enhancements or research extensions.

6. Revised Project Plan

7. References

[1] X. Wang, L. Xie, C. Dong, and Y. Shan, "Real-ESRGAN: Enhanced Super-Resolution Generative Adversarial Networks," arXiv preprint arXiv:2107.10833, 2021. [Online]. Available: <https://arxiv.org/abs/2107.10833>

- [2] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Whisper: Robust Speech Recognition via Large-Scale Weak Supervision,” OpenAI, 2022. [Online]. Available: <https://openai.com/research/whisper>
- [3] MyHeritage, “Deep Nostalgia — Animate Faces in Old Photos,” 2021. [Online]. Available: <https://www.myheritage.com/deep-nostalgia>
- [4] ElevenLabs, “AI Speech Software for Realistic Voice Generation,” 2023. [Online]. Available: <https://elevenlabs.io>
- [5] Descript, “AI Voice Cloning and Audio Editing Platform,” 2023. [Online]. Available: <https://www.descript.com>
- [6] Alzheimer’s Disease International, World Alzheimer Report 2022: Life after Diagnosis — Navigating Treatment, Care, and Support, 2022. [Online]. Available: <https://www.alzint.org/resource/world-alzheimer-report-2022>
- [7] OpenAI, GPT-4 Technical Report, 2023. [Online]. Available: <https://openai.com/research/gpt-4>

Appendix A: Glossary

Appendix B: IV & V Report

(Independent verification & validation)
IV & V Resource

Name

Signature

S#	Defect Description	Origin Stage	Status	Fix Time	
				Hours	Minutes
1					
2					
3					
...					

Table 3: List of non-trivial defects